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Continual Learning with Probabilistic Mixture of Experts via Variational Moment Propagation

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***Abstract*—In exemplar-free continual learning (CL), the learning agent is constrained from storing or revisiting samples from earlier tasks. As a result, the model is highly vulnerable to catastrophic forgetting, leading to significant deterioration in performance on previously learned tasks. Most existing CL models are deterministic, producing single-point estimates while neglecting underlying uncertainty, which limits their reliability in real-world deployment. Although uncertainty modeling can play a crucial role in improving continual learning—by guiding adaptation and mitigating forgetting—such approaches remain insufficiently explored, particularly in mixture-of-experts frameworks. In this paper, we develop and implement a Probabilistic Mixture of Experts (ProMoE) model that is well-suited for continual learning. Unlike conventional probabilistic methods that estimate uncertainty via posterior sampling, our framework directly propagates the first two moments of the variational posterior distribution through expert architecture. Specifically, we approximate the propagation of these moments through nonlinear layers using a Taylor series expansion, with variational variance providing parameters' uncertainty. The inherent regularization in variational inference significantly helps mitigate catastrophic forgetting, even without replaying samples from previous tasks. We empirically evaluate the performance of the proposed approach on various continual learning benchmarks and demonstrate the improved accuracy and reduced forgetting compared to existing approaches in the literature.**

***Index Terms*—Continual Learning, Catastrophic Forgetting, Mixture of Experts, Moment Propagation**

I. Introduction

Continual learning (CL), also known as lifelong or incremental learning, enables models to learn sequential tasks while balancing adaptability (plasticity) and retention of prior knowledge (stability). However, conventional neural networks suffer from *catastrophic forgetting*, a phenomenon in which learning new tasks leads to substantial degradation in performance on prior tasks [1]–[4]. A straightforward solution to mitigate forgetting involves replaying instances from earlier tasks during training on new tasks. However, reply methods introduce challenges related to sample selection, data privacy, increased storage requirements, and additional computational overhead [5], [6]. While deterministic continual learning methods have achieved substantial progress, the probabilistic or Bayesian paradigm remains comparatively underexplored. This limited attention largely stems from concerns about the scalability of probabilistic models to large datasets and complex architectures. Nonetheless, Bayesian neural networks (BNNs) offer several key advantages, including robustness, uncertainty-awareness, and well-calibrated inference. These are particularly well-suited and required for continual learning in real-world applications, such as medical diagnostics, environmental sensing and monitoring, and self-driving cars.

The advantages of modeling the learning agent with a probabilistic neural network are threefold. First, probabilistic models inherently represent *epistemic uncertainty* by maintaining a distribution over weights rather than fixed point estimates [7], [8]. Weight distribution propagates through the network and enables quantifying confidence in predictions, facilitating robust adaptation when encountering new tasks or data distributions. Second, Bayesian inference offers a natural regularization mechanism based on the Kullback–Leibler (KL) divergence, constrains parameter drift across tasks, and supports continual learning [9]. This mitigates catastrophic forgetting without relying on explicit replay buffers or generative replay of samples from previous tasks. Third, Bayesian approaches provide calibrated inference, enabling the model to align task boundaries, prevent overfitting to specific tasks, and maintain stability during sequential updates [10].

Despite these advantages, existing probabilistic continual learning methods still rely on replay-based strategies and are restricted to relatively small model architectures. Moreover, many Bayesian model formulations significantly increase computational overhead, either by substantially expanding the number of parameters or by relying on costly sampling-based inference. In contrast, our proposed framework leverages moment propagation to achieve uncertainty modeling with negligible computational overhead and without relying on parameter sampling. For example, when implementing a ResNet architecture [11], our probabilistic model requires approximately 467K parameters—nearly identical to its deterministic counterpart (466K), representing only a 0.27% increase. Whereas a conventional Bayesian formulation of the same architecture requires 946K parameters [12], demonstrating the substantially lower parameter overhead of our approach.

In this paper, we propose an exemplar-free continual learning approach based on a probabilistic mixture-of-experts (ProMoE) that mitigates catastrophic forgetting while balancing stability and plasticity. Our results show that properly designed Bayesian models can scale effectively, leverage uncertainty, and achieve strong CL performance without replay buffers.

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The proposed contributions can be summarized as follows:

- Develop a probabilistic mixture-of-experts (ProMoE) tailored for continual learning, where uncertainty-aware experts and gating mechanisms enable adaptive task specialization, controlled knowledge transfer, and mitigation of catastrophic forgetting across sequential tasks.
- Introduce a moment-propagation-based probabilistic formulation within the mixture-of-experts framework that propagates the first two moments of the variational posterior through expert networks, allowing scalable uncertainty estimation without significant parameter overhead.
- Demonstrate how ProMoE leverages uncertainty to balance stability and plasticity, achieving strong continual learning performance, reduced forgetting, and improved generalization without reliance on replay buffers.
- Evaluate the proposed framework on standard continual learning benchmarks and demonstrate competitive or superior performance compared to existing Bayesian methods, with improved scalability and reduced computational overhead in exemplar-free settings.

II. Related Works

A. Continual Learning

Existing approaches to continual learning can be broadly categorized into three groups. (1) *Replay-based* methods [13]–[16] alleviate forgetting by retaining a limited set of representative samples from previous tasks, which are replayed during training on subsequent tasks. (2) *Regularization-based* methods reduce interference across tasks by constraining updates to parameters that are considered important for previously learned tasks, typically through additional regularization terms [3], [17]–[19]. (3) *Dynamic or expandable* architectures combat forgetting by adapting the model structure over time, for instance by expanding the network to incorporate new tasks, pruning less relevant components, or isolating task-specific parameters [20]–[23]. Within this taxonomy, our probabilistic approach falls into the regularization-architecture category, combining variational uncertainty-based regularization with a Mixture-of-Experts framework to preserve knowledge across tasks (details in Section III).

B. Probabilistic or Bayesian Continual Learning

Despite the well-established advantages of Bayesian neural networks (BNNs) in enhancing model generalization, their application to continual learning (CL) has remained relatively limited [8], [9], [24], [25]. Nguyen *et al.* introduced Variational Continual Learning (VCL), which employs variational inference and propagates samples from the variational posterior to model uncertainty [24]. Ebrahimi *et al.* modified VCL by leveraging the uncertainty of model weights from earlier tasks to adapt task-specific learning rates for new tasks [25]. Ahn *et al.* further extended the VCL framework by incorporating per-weight regularization and modeling uncertainty at the node level [9]. Kumar *et al.* proposed a Bayesian CL approach based on Indian Buffet Process priors (IBPCL), which learns task-specific masks to identify relevant parameters and facilitate inter-task knowledge transfer [26]. Hihn and Braun introduced Hierarchical Variational Continual Learning (HCVL), where each layer comprises a mixture of sub-networks whose Bayesian posteriors are aligned through hierarchical matching [27]. However, HCVL was restricted to generative modeling and reinforcement learning, with a heavy reliance on replay buffers and simultaneous training of all sub-networks [27].

Li *et al.* studied MoE for continual learning in linear synthetic settings, lacking probabilistic uncertainty modeling and validation on deep or real-world tasks [28]. Overall, existing probabilistic approaches to continual learning often depend on replay-based mechanisms and lack compelling evidence of effectiveness in exemplar-free settings or scalability to complex, large-scale networks. In contrast, we propose a modular framework that addresses these shortcomings: it supports incremental learning without requiring a replay buffer, and demonstrates strong empirical performance on large-scale probabilistic MoE architectures and challenging benchmarks.

III. Preliminaries on Variational Inference

The Bayesian theory of learning provides a principled framework for modeling uncertainty by treating model parameters as random variables with probability distributions [7]. Given observed data $\mathcal{D}$ and parameters prior $p(\theta)$, Bayes' theorem computes the posterior distribution, $p(\theta|\mathcal{D})$, as follows:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \tag{1}$$

where $p(\mathcal{D}|\theta)$ is the likelihood, and $p(\mathcal{D})$ is the marginal likelihood (or evidence). The posterior reflects updated beliefs about the parameters after observing data. In neural networks, computing the exact posterior is intractable due to the nonlinearity and high dimensionality of the parameter space. Variational Inference (VI) [29] approximates the true posterior $p(\theta|\mathcal{D})$ with a simpler distribution $q(\theta)$, often chosen from a family of Gaussian distributions. The goal is to minimize the Kullback-Leibler (KL) divergence between the approximate and true posterior distributions. This is achieved by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(\theta) = \mathbb{E}_{q(\theta)}[\log p(\mathcal{D} \mid \theta)] - \text{KL}(q(\theta) \,\|\, p(\theta)). \tag{2}$$

The first term assigns high probability to parameters that explain the data well, while the second term regularizes the distribution to stay close to the prior. Variational inference thus turns posterior approximation into an optimization, providing a scalable and efficient approach for Bayesian learning, particularly well-suited for deep neural networks.

IV. Proposed Probabilistic Mixture-of-Expert for Continual Learning

In this section, we formally describe the proposed probabilistic MoE model based on variational moment propagation. We begin by formulating the CL problem, followed by the design and implementation of the variational moment propagation layers. Finally, we present the probabilistic MoE architecture for continual learning built upon these layers.

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A. Notations and Problem Setup

Suppose a continual learning model aims to learn from a sequence of tasks $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_T\}_{t=1}^{T}$ in a sequential manner, where each task is encountered only once, and previous data are not accessible during the training on subsequent tasks. The model is required to accumulate knowledge over time while retaining information from earlier tasks. Each task $\mathcal{T}_t$ is associated with a dataset $\mathcal{D}_t = \{(\mathbf{X}_i^{(t)}, \mathbf{y}_i^{(t)})\}_{i=1}^{N_t}$, where $(\mathbf{X}_i^{(t)}, \mathbf{y}_i^{(t)})$ are sampled from an underlying data distribution in task t. We assume that the tasks are *disjoint*, such that their label spaces do not overlap, i.e., $\mathcal{Y}_i \cap \mathcal{Y}_j = \emptyset, \quad \forall i \neq j$. In incremental learning, a model seeks to learn a single unified hypothesis f_θ that maintains performance across the sequence of tasks drawn from distinct data-generating distributions $\{P_t(\mathbf{X}, \mathbf{y})\}_{t=1}^{T}$. The learning objective is therefore expressed as the minimization of the aggregate expected risk:

$$\theta^\star = \arg\min_\theta \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim P_t(\mathbf{X},\mathbf{y})} \big[\mathcal{L}(f_\theta(\mathbf{x}), \mathbf{y})\big], \quad (3)$$

where the goal is to derive a parameterization that achieves consistent generalization across all tasks under non-stationary data distributions. $\mathcal{L}(\cdot)$ denotes a task-specific loss function. The challenge in this setting lies in preventing catastrophic forgetting, where learning new tasks adversely affects performance on previously learned ones.

B. Probabilistic Feature Extraction (ProFE)

Let the input to the expert network be $\mathbf{X} \in \mathbb{R}^{H\times W\times C_{\text{in}}}$. Each expert comprises multiple convolutional blocks for feature extraction, each with M convolutional kernels. Each mth kernel is denoted by $\mathbf{K}^{(m)} \in \mathbb{R}^{K_h\times K_w\times C_{\text{in}}}$. We define independent Gaussian priors over these probabilistic kernels, and thus the posterior is also Gaussian, i.e., $\text{vec}(\mathbf{K}^{(m)}) \sim \mathcal{N}(\boldsymbol{\mu}_m, \sigma_m^2\mathbf{I})$, where vec is the vectorization operation, $\boldsymbol{\mu}_m$ is the vectorized kernel mean and σ_m^2 its variance. Assuming independence across kernels facilitates the extraction of decorrelated and complementary features. The learnable probabilistic kernels of all convolution blocks the expert networks are parameterized by $\boldsymbol{\Theta}_{\text{Conv}} = \{\boldsymbol{\mu}_m, \sigma_m^2\}_{m=1}^{M}$, and are optimized by maximizing the ELBO objective during training.

The input image or feature map $\mathbf{X}$ is also modeled as a diagonal Gaussian random variable with mean $\mathbf{M}_X$ and variance $\mathbf{S}_X$, both of dimension $\mathbb{R}^{H\times W\times C_{\text{in}}}$. We assume the input distribution to be independent of the kernels' distributions. To propagate variational moments (mean and variance) through the feature extraction (convolutional) blocks, we extract overlapping local patches $\mathbf{M}_X^{(i,j)}$ and $\mathbf{S}_X^{(i,j)}$ from the mean and variance of the input, $\mathbf{X}$, each having the same dimensions as $\mathbf{K}^{(m)}$. These patches are vectorized and stacked row-wise to form matrices $\mathbf{X}_\mu, \mathbf{X}_\sigma \in \mathbb{R}^{H'W'\times K_hK_wC_{\text{in}}}$, where $H'\times W'$ are the spatial dimensions of the resulting feature map, determined by the convolution stride and padding. Similarly, we define $\mathbf{K}_\mu = [\,\boldsymbol{\mu}_1 \cdots \boldsymbol{\mu}_M\,] \in \mathbb{R}^{K_hK_wC_{\text{in}}\times M}$ and $\boldsymbol{\sigma}^\top = [\,\sigma_1^2 \cdots \sigma_M^2\,]$ as the stacked kernels mean and variance, respectively.

Let $\mathcal{F}$ denote the output of a convolutional block after vectorizing spatial dimensions, $\mathcal{F} \in \mathbb{R}^{H'W'\times M}$. The first- and second-order moments of $\mathcal{F}$ can be efficiently derived as:

$$\begin{aligned} \mathcal{F}_\mu &= \mathbf{X}_\mu\mathbf{K}_\mu, \quad (4)\\ \mathcal{F}_\sigma &= \mathbf{X}_\sigma(\mathbf{K}_\mu \odot \mathbf{K}_\mu) + \|\mathbf{X}_\mu\|_{\text{row}}^2 \cdot \boldsymbol{\sigma}^\top + \left(\sum_{p=1}^{K_hK_wC_{\text{in}}} \mathbf{X}_{\sigma,p}\right) \cdot \boldsymbol{\sigma}^\top, \end{aligned}$$

where $\odot$ denotes element-wise (Hadamard) multiplication and $\|\mathbf{X}_\mu\|_{\text{row}}^2$ is the vector of squared ℓ_2-norms of each row of $\mathbf{X}_\mu$: $\|\mathbf{X}_\mu\|_{\text{row}}^2 = [\,\|\mathbf{X}_{\mu,1}\|_2^2, \ldots, \|\mathbf{X}_{\mu,n}\|_2^2\,]$. For simplicity and without loss of generality, we assume the input image, $\mathbf{X}$, to the first convolutional block only, to be deterministic (i.e., $\mathbf{X}_\sigma = 0$). Thus, Eq. 4 simplifies to:

$$\mathcal{F}_\mu = \mathbf{X}_\mu\mathbf{K}_\mu, \qquad \mathcal{F}_\sigma = \|\mathbf{X}_\mu\|_{\text{row}}^2 \cdot \boldsymbol{\sigma}^\top. \quad (5)$$

This formulation enables tractable moment propagation through convolutional feature extraction while preserving probabilistic independence among kernels, resulting in efficient and uncertainty-aware feature learning.

C. Probabilistic Classifier (ProbFC)

Let $\mathbf{U}$ denote the weight matrix of a probabilistic linear layer, where each column $\mathbf{u}^{(r)}$ represents the weights associated with neuron r. We assume a Gaussian variational posterior for each neuron's weight vector: $\mathbf{u}^{(r)} \sim \mathcal{N}(\boldsymbol{\mu}_r, \sigma_r^2\mathbf{I})$, where $\boldsymbol{\mu}_r$ and σ_r^2 denote the mean and variance of the variational posterior, respectively. We construct the mean weight matrix and variance vector as $\mathbf{U}_\mu = [\,\boldsymbol{\mu}_1 \cdots \boldsymbol{\mu}_R\,]$ and $\boldsymbol{\sigma}^\top = [\,\sigma_1^2 \cdots \sigma_R^2\,]$, where R represents the number of neurons (i.e., output dimension) in the layer. The variational parameters of the probabilistic linear fully-connected layer are then given by $\boldsymbol{\Theta}_{\text{FC}} = \{\boldsymbol{\mu}_r, \sigma_r^2\}_{r=1}^{R}$, which are optimized by maximizing the ELBO objective during training. Given an input probabilistic feature matrix $\mathbf{X}$ with mean $\mathbf{M}_X$ and variance $\mathbf{S}_X$, the first- and second-order moments of the output $\mathcal{F}$ after a probabilistic linear layer can be computed analogously to the convolutional propagation described in Eq. 4. Specifically, the output mean and variance are:

$$\begin{aligned} \mathcal{F}_\mu &= \mathbf{M}_X\mathbf{U}_\mu, \quad (6)\\ \mathcal{F}_\sigma &= \mathbf{S}_X(\mathbf{U}_\mu \odot \mathbf{U}_\mu) + \|\mathbf{M}_X\|_{\text{row}}^2 \cdot \boldsymbol{\sigma}^\top + \left(\sum_{j=1}^{D_{\text{in}}} \mathbf{S}_{X,j}\right) \cdot \boldsymbol{\sigma}^\top, \end{aligned}$$

where D_{in} denotes the input dimension of the linear layer.

D. Probabilistic Activation

Activation functions—such as sigmoid, hyperbolic tangent, rectified linear unit, and their variants—introduce nonlinear transformations that make the exact propagation of uncertainty analytically intractable. To obtain a tractable formulation for propagating the mean and variance through such nonlinearities, we employ a first-order approximation similar to the delta method or the extended Kalman filter. Let g denote an activation function applied to the output of a convolutional or fully-connected layer, such that $\tilde{\mathcal{F}} = g(\mathcal{F})$. The mean and

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variance of the transformed variable $\tilde{\mathcal{F}}$, denoted by $\tilde{\mathcal{F}}_\mu$ and $\tilde{\mathcal{F}}_\sigma$, can be approximated as:

$$\tilde{\mathcal{F}}_\mu = g(\mathcal{F}_\mu), \qquad \tilde{\mathcal{F}}_\sigma = (\nabla g(\mathcal{F}_\mu))^{\circ 2} \odot \mathcal{F}_\sigma, \tag{7}$$

where $\nabla g(\mathcal{F}_\mu)$ is the gradient of g with respect to $\mathcal{F}$, evaluated at its mean $\mathcal{F}_\mu$, and $(\cdot)^{\circ 2}$ denotes the element-wise square.

E. Probabilistic Mixture of Experts (ProMoE)

The proposed probabilistic mixture of experts (ProMoE) consists of K probabilistic deep neural networks (named experts). Each expert architecture includes probabilistic feature extraction (ProFE), referred to as $\mathcal{B}_k$, where $k = 1, \ldots, K$, and linear fully-connected layers (ProFC) for classification.

When training each expert network on any given task, we approximate the posterior distribution over the parameters using variational inference. The optimization problem is formulated as minimizing the negative evidence lower bound (ELBO loss in Eq. 2). We propagate moments of the variational posterior distribution, i.e., $\theta = \{\boldsymbol{\Theta}_{\text{Conv}}, \boldsymbol{\Theta}_{\text{FC}}\}$, as explained in Sections IV-B - IV-D. The posterior variance represents the uncertainty in the experts' parameters. This variance, when propagated to the output, reflects the uncertainty over the predictions of each expert.

Each expert is trained on a single incremental task for the first K tasks. Then, we define a gating mechanism to select a single expert (from the K pool) and fine-tune it on a single task for the $t > K$ tasks. Unlike deterministic models that result in a single feature embedding (typically mean) from the expert ProFE, our formulation generates a distribution over the latent representations. It is also important to note that the variance is learned alongside the mean, whereas existing Bayesian methods rely on ad hoc approximations for variance. Specifically, the expert's ProFE generates a moment pair $(\mathcal{F}_\mu^t, \mathcal{F}_\sigma^t)$, representing the mean and variance of the latent feature distribution given the input $\mathbf{X}$, i.e., $(\mathcal{F}_\mu^t, \mathcal{F}_\sigma^t) = \mathcal{B}_t(\mathbf{X})$. Then, the feature moments are passed through the fully-connected layers to obtain the probabilistic logits such that: $(\mathcal{F}_\mu^t, \mathcal{F}_\sigma^t) \leftarrow \text{ProbFC}(\mathcal{F}_\mu^t, \mathcal{F}_\sigma^t)$.

In the context of multi-class classification, the likelihood $p(\mathcal{D}|\theta)$ is modeled by a multinomial distribution. The expectation $\mathbb{E}_{q(\theta)}[.]$ is approximated via a Monte Carlo estimator by perturbing the logit means, i.e., $\mathcal{F}^t = \mathcal{F}_\mu^{(t)} + \mathcal{F}_\sigma^{(t)} \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. Sampling is confined to the loss estimator; the forward propagation of the posterior means and variances is performed in closed form (completely sample-free propagation). Thus, the expected negative log-likelihood (NLL) term in Eq. 2 is equivalent to the cross-entropy loss computed between the ground-truth labels and the predicted class probabilities derived from the softmax function.

To promote stable knowledge transfer across sequential tasks, we employ an *uncertainty-aware feature-based* knowledge distillation strategy between the current and previous probabilistic feature embeddings. This formulation aligns the current expert's stochastic embeddings, $\mathcal{F}^t$, with the prior task's uncertainty distribution via $\mathcal{F}^{t-1}$, enabling smooth knowledge distillation and effective retention of previously learned knowledge. Thus, we minimize the Euclidean distance between these two stochastic embeddings as:

$$\mathcal{L}_{\text{KD}} = \|\mathcal{F}^t - \mathcal{F}^{t-1}\|_2^2. \tag{8}$$

The overall optimization objective function of the framework at task t can be expressed as:

$$\mathcal{L}(\theta, \mathcal{D}_t) = (1-\alpha)\Big[\mathcal{L}_{\text{NLL}} + \gamma\, \text{KL}\big[q_\phi(\theta) \,\|\, p(\theta)\big]\Big] + \alpha\, \mathcal{L}_{\text{KD}}, \tag{9}$$

where α and γ are hyperparameters that regulate the strength of the regularization and distillation terms.

Summary of ProMoE: In the ProMoE framework, the first K tasks are individually assigned to distinct experts within the mixture and optimized through the ELBO loss (Eq. 2), ensuring principled probabilistic learning of task-specific posteriors. For subsequent tasks ($t > K$), a single optimal expert is selected from the pool of experts and fine-tuned using the hybrid objective (Eq. 9), combining Bayesian regularization with stochastic feature distillation.

For selecting the optimal expert, we model expert features of each class c as a Gaussian distribution parameterized by a mean vector $\boldsymbol{\mu}_{c,k}$ and a covariance matrix $\boldsymbol{\Sigma}_{c,k}$, both estimated from the class-wise latent feature means $\mathcal{F}_\mu$. The resulting Gaussian mixture model for expert k is defined as $\mathcal{G}_{c,k} = \mathcal{N}(\boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k})$. We then select the expert whose learned class distributions exhibit the maximum overlap with the classes of the previous task. To quantify this overlap, we employ a gating metric based on the Jensen–Shannon (JS) divergence [30], computed between the feature distributions of the classes in the current and previous tasks. The expert that yields the highest JS divergence score is selected and subsequently fine-tuned for the new task. This strategy enables efficient adaptation to new tasks while preserving previously acquired knowledge and maintaining consistent uncertainty propagation through each layer of each expert. We present the overall continual learning framework in Algorithm 1.

V. Experiments and Results

A. Datasets and Continual Learning Protocols

We evaluate the proposed model on a diverse set of widely used image classification benchmarks that span varying levels of visual complexity, ranging from simple grayscale images to challenging natural images. The datasets include MNIST, Fashion-MNIST (FMNIST), CIFAR-10, CIFAR-100, TinyImageNet-200, and Permuted MNIST (PMNIST). MNIST consists of 28×28 grayscale images of handwritten digits (0–9) [31]. Fashion-MNIST contains 28×28 grayscale images of clothing and fashion-related items such as T-shirts, trousers, dresses, and bags, exhibiting higher intra-class variability and more complex textures than MNIST [32]. CIFAR-10 and CIFAR-100 are natural image datasets composed of 32×32 RGB images. CIFAR-10 contains 10 coarse-grained object categories, including animals and vehicles, while CIFAR-100 comprises 100 fine-grained categories with increased visual diversity and classification difficulty [33]. TinyImageNet-200 further increases dataset complexity by providing 200 classes

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Algorithm 1 ProMoE: Continual Training and Fine-Tuning

Require: Datasets $\{\mathcal{D}_t\}_{t=1}^{T}$, experts $\{\mathcal{B}_k(\theta)\}_{k=1}^{K}$, and hyper-parameters α, γ, η

for each task $t = 1, 2, \ldots, T$ **do**
  Sample minibatch $(x, y) \sim \mathcal{D}_t$
  if $t \leq K$ **then** ▷ Training on initial experts
    Forward pass $(\mathcal{F}_\mu, \mathcal{F}_\sigma) \leftarrow \mathcal{B}_t(x)$
  else ▷ Fine-tuning
    Select expert $\mathcal{B}_{\hat{k}}$
    Forward pass $(\mathcal{F}_\mu, \mathcal{F}_\sigma) \leftarrow \mathcal{B}_{\hat{k}}(x)$
  end if
  Compute logits $(\mathcal{F}_\mu, \mathcal{F}_\sigma) \leftarrow \mathrm{ProbFC}(\mathcal{F}_\mu, \mathcal{F}_\sigma)$
  Reparameterize $\hat{y} \leftarrow \mathcal{F}_\mu + \mathcal{F}_\sigma \odot \epsilon, \epsilon \sim \mathcal{N}(0, I)$
  $\mathcal{L}_{\mathrm{NLL}} \leftarrow \mathrm{NLL}(y, \hat{y})$
  if $t \leq K$ **then**
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{NLL}} + \gamma\, \mathrm{KL}[q_\phi(\theta) \,\|\, p(\theta)]$
  else
    $\mathcal{L} \leftarrow (1-\alpha)[\mathcal{L}_{\mathrm{NLL}} + \gamma\, \mathrm{KL}[q_\phi(\theta) \,\|\, p(\theta)]] + \alpha\, \mathcal{L}_{\mathrm{KD}}$
  end if
  Optimize $\theta \leftarrow \theta - \eta\, \nabla_\theta \mathcal{L}$
end for

of natural images consisting of 64×64 pixels, spanning animals, objects, and scenes, with higher semantic and visual variability. Permuted MNIST (PMNIST) is a domain-incremental learning benchmark derived from MNIST, where pixel locations are randomly permuted using different random seeds to create multiple visually distinct domains while preserving the original digit labels.

To construct task sequences for continual learning, we follow standard learning protocols. MNIST, Fashion-MNIST, and CIFAR-10 are each partitioned into five tasks, with two classes per task. CIFAR-100 is divided into 10 and 20 tasks, respectively, each containing an equal number of classes per task (10 and 5 classes in each task). For TinyImageNet, we construct 10 sequential task having 20 classes in each task. For PMNIST, we generate 10 different domains using 10 distinct random seeds, where each permutation defines a separate task while sharing the same label space across tasks. This experimental design allows us to systematically evaluate the model's ability to learn sequentially across datasets of increasing visual complexity.

B. Compared Baselines

We compare the proposed ProMoE against a set of strong and representative probabilistic continual learning baselines, including Elastic Weight Consolidation (EWC) [3], Hard Attention to Task (HAT) [34], Variational Continual Learning (VCL) [24], Uncertainty-guided Continual Learning (UCB) [25], Uncertainty-based Continual Learning (UCL) [9], Bayesian Structural Adaptation for Continual Learning (IBPCL) [26], and Functional Space Variational Inference (FSVI) [35].

C. Implementation Details

All experiments are conducted under the FACIL framework [5]. We implement a probabilistic ResNet-18 feature extractor (architecture adopted from [36]), while we derive and train the moment propagation layers from scratch. We use a stochastic gradient descent (SGD) optimizer, and set the initial learning rate, η, to 0.05 for 200 epochs. We employ the PyTorch machine learning library to implement the proposed framework [37]. We use NVIDIA RTX^{TM} A6000 Ada Gen with 48 GB of RAM to run all of our experiments.

D. Evaluation Metric

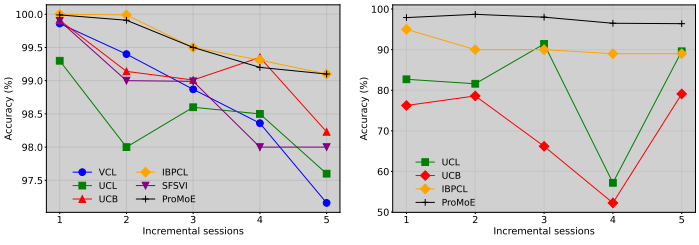
We denote the model's accuracy after each t-th task as A_t. The average performance across all incremental tasks (Avg) is defined as:

$$\mathrm{Avg} = \frac{1}{T} \sum_{t=1}^{T} \mathrm{A}_t. \tag{10}$$

E. Results

In Figure 1, we illustrate the average incremental accuracy measured after training on each task for ProMoE compared to other baselines. The plots show how the model accumulates past knowledge while learning new ones. Table I provides a comprehensive comparison of continual learning (CL) baselines that employ probabilistic/Bayesian models. From these results, ProMoE with probabilistic feature extractors consistently outperform existing Bayesian or uncertainty-based CL approaches. For lower-complexity datasets such as MNIST, FMNIST, and PMNIST, PropMoE performs on par with or slightly surpasses comparable baselines. Specifically, ProMoE attains an accuracy of 99.50% on MNIST, closely matching methods like IBPCL [26]. Similarly, for FMNIST, ProMoE achieves 99.13%, which is the highest among all other methods and is approximately 1.13% higher than the second-best method, IBPCL [26]. We can observe a similar pattern for Permuted MNIST.

As task difficulty increases, ProMoE exhibits markedly improved robustness and accuracy, even *without replaying* samples from previous tasks. On CIFAR-10, ProMoE achieves 97.60%, outperforming the second-best method IBPCL [26] by approximately 5%. This difference in accuracy increases



(a) MNIST split into 5 tasks. (b) CIFAR-10 with 5 tasks.

Fig. 1: Average incremental accuracy of the proposed ProMoE measured after each task for different scenarios: Split MNIST (left) and CIFAR-10 (right). ProMoE consistently demonstrates higher accuracy (better retention of previously learned knowledge) than other Bayesian CL approaches.

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TABLE I: Average incremental accuracy (%) for Bayesian CL Models with different numbers of incremental tasks evaluated on MNIST (T=5), FMNIST (T=5), CIFAR-10 (T=5), PMNIST (T=10), CIFAR-100 (T=10, 20), and TinyImageNet (T=10). Each task contains an equal number of classes. Best results are in bold. Second-best methods are underlined. Higher is better.

Approach	Replay	MNIST	FMNIST	CIFAR-10	PMNIST	CIFAR-100		TinyImageNet
		Avg ↑	Avg ↑	Avg ↑	Avg ↑	Avg ↑	Avg ↑	Avg ↑
EWC [3]	✓	99.01	97.85	81.85	96.86	62.54	65.20	42.30
VCL [24]	✓	97.80	97.55	81.89	97.11	64.79	72.46	40.07
UCL [9]	✓	98.97	98.00	85.11	96.55	64.81	74.01	46.46
HAT [34]	✓	99.40	98.00	92.11	96.99	72.81	75.01	44.46
UCB [25]	✓	99.30	97.21	70.48	96.55	65.22	79.44	–
IBPCL [26]	✓	**99.58**	98.00	90.53	**97.53**	68.84	73.13	48.46
FSVI [35]	✓	98.00	97.98	–	97.22	–	–	–
ProMoE (ours)	×	99.50	**99.13**	**97.60**	97.32	**81.51**	**85.02**	**53.12**

TABLE II: Average incremental accuracy (%) on the CIFAR-100 dataset with ten sequential tasks. Each cell reports the average classification accuracy on the corresponding task $\mathcal{T}_i$ after training up to that session. Each task consists of an equal number of classes.

Sessions	$\mathcal{T}_1$	$\mathcal{T}_2$	$\mathcal{T}_3$	$\mathcal{T}_4$	$\mathcal{T}_5$	$\mathcal{T}_6$	$\mathcal{T}_7$	$\mathcal{T}_8$	$\mathcal{T}_9$	$\mathcal{T}_{10}$	Avg.
$\mathcal{T}_1$	85.2	–	–	–	–	–	–	–	–	–	85.2
$\mathcal{T}_2$	85.2	73.7	–	–	–	–	–	–	–	–	79.5
$\mathcal{T}_3$	85.2	73.7	88.7	–	–	–	–	–	–	–	82.5
$\mathcal{T}_4$	85.2	73.7	88.7	80.9	–	–	–	–	–	–	82.1
$\mathcal{T}_5$	85.2	73.7	88.7	80.9	89.5	–	–	–	–	–	83.6
$\mathcal{T}_6$	85.2	73.7	88.7	80.9	89.3	71.3	–	–	–	–	81.5
$\mathcal{T}_7$	85.2	73.7	88.7	80.9	89.5	72.7	79.0	–	–	–	81.4
$\mathcal{T}_8$	85.2	73.7	88.7	80.9	88.9	72.5	78.9	76.9	–	–	80.7
$\mathcal{T}_9$	83.9	73.7	89.3	81.3	88.4	73.4	79.0	77.0	78.8	–	80.5
$\mathcal{T}_{10}$	81.9	72.7	88.9	81.0	88.7	72.5	78.4	76.8	79.0	76.0	79.6
										Avg	**81.7**

further when evaluating on more complex datasets such as CIFAR-100 and TinyImageNet. For example, on CIFAR-100 with 10 tasks, the model achieves 81.51%, which is 8.7% higher than the second-best method, HAT [34]. A similar trend is observed for TinyImageNet, confirming ProMoE's scalability and resilience on more complex visual benchmarks. The methods marked with $(-)$ in Table I are not included due to the lack of implementations in the original publication.

In Table II, we report the detailed accuracy for the ten-task continual learning experiment on the CIFAR-100 dataset. The model starts with 85.2% accuracy on $\mathcal{T}_1$ and maintains strong performance as new tasks are introduced continually, with stage-wise averages ranging from 79.5% to 83.6%. This stability across tasks indicates effective knowledge consolidation and moderate resistance to forgetting. Even after all ten tasks, the model retains a high average accuracy of 79.6%, suggesting that previously learned representations remain robust. The final average incremental accuracy (Avg) of 81.7% confirms that the model achieves a favorable trade-off between plasticity and stability, continually integrating new visual concepts while preserving prior task performance.

VI. Conclusions

In this work, we introduce a probabilistic mixture-of-experts (ProMoE) architecture based on variational moment-propagation for continual learning. The proposed framework employs variational inference with Evidence Lower Bound (ELBO) optimization, in which the first two moments of the variational distribution are propagated across MoE layers using a Taylor series approximation. This formulation enables efficient incorporation of uncertainty and adaptive knowledge transfer to future tasks. The KL-divergence term in the ELBO loss provides an inherent and principled form of regularization that mitigates catastrophic forgetting. Unlike conventional approaches that rely on replay buffers or stored exemplars, our method achieves continual adaptation without revisiting past data, thereby preserving scalability and privacy. Moreover, the proposed approximation introduces only a minimal parameter

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overhead compared to conventional Bayesian formulations, making it suitable for larger architectures. Experiments on standard continual learning benchmark datasets demonstrate the robustness and scalability of the proposed ProMoE framework. Overall, the results show that our approach effectively alleviates catastrophic forgetting while maintaining strong generalization, establishing a memory-free, parameter-efficient, and Bayesian-consistent paradigm for lifelong learning.

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